Credit Risk Portfolio Analytics -- Portfolio monitoring, modelling & analyst review

Portfolio: 14,685 loan accounts | 10,000 customers | 16.10% default rate | 2,364 defaults | $601.1M original exposure

Dashboard: analyst workflow

- **Portfolio overview:** Filter by loan type, region, employment and default status; monitor default rate, account counts and exposure.
- **Risk segmentation:** Compare default rates by loan type, income, employment, age and region; use the income–employment view to explore combined risk segments.
- **Account explorer:** Review borrower and loan details, observed outcome, estimated default probability & main model drivers.
- **Anomaly review:** Inspect flagged accounts, reasons and repayment metrics; record Confirmed/Dismissed decisions.
- **Ask the Portfolio:** Ask supported questions about the filtered data; verify answers against the displayed source and chart.

Mandatory AI features

Ask the Portfolio (LLM): Analyst submits a natural-language question. The LLM (Gemini) maps it to a supported query; figures are calculated from dashboard data, and the answer includes its source. Unsupported questions are declined. False positive: question mapped to the wrong metric or segment. False negative: valid question declined. Human review: verify the answer against its displayed source and chart.
Anomaly detection: Flags accounts with unusual repayment or bureau patterns relative to their own history or peer segment. Each flag shows its reason and supporting metrics. False positive: unusual behaviour has a legitimate explanation. False negative: an anomaly is not flagged. Human review: analyst examines the evidence and records Confirmed or Dismissed.

Modelling approach & results

1. **Load and clean:** Profile the five source tables, handle missing values and duplicates, aggregate bureau, repayment and collection records, then join them at account level. Validate the target definition and final dataset grain.
2. **Engineer features:** Create affordability ratios (loan_to_income_ratio, emi_payment_ratio), repayment signals (late/missed payments and average DPD), and bureau risk features (average DPD, active-account ratio and outstanding balance per active account).
3. **Split and preprocess:** Split by customer_id to prevent customer overlap; use grouped cross-validation. Fit imputers, scaling and categorical encoding within each training fold to avoid leakage.
4. **Train and tune:** Compare Logistic Regression and XGBoost using GridSearchCV and grouped CV, scoring with ROC-AUC and PR-AUC. Compare class weighting with 20% random oversampling applied only to training folds.
5. **Select threshold and features:** Choose the decision threshold using out-of-fold predictions. Assess permutation feature importance and test removing low-importance features with grouped CV.
6. **Evaluate:** Freeze the selected model and threshold, then report held-out ROC-AUC, KS and Gini, along with classification metrics at the selected threshold. Logistic Regression was selected based on stronger grouped CV performance.

Held-out results (Logistic Regression — chosen model): ROC-AUC 0.6881 | KS 0.2728 | Gini 0.3763
(Gini = 2 × ROC-AUC − 1)

Top portfolio insights

1. **Mortgages:** 52.31% of original exposure ($314.47M) and a 24.65% default rate. **So what:** Prioritize mortgage reviews of underwriting, affordability and repayment performance.
2. **Income:** Defaults decline as income rises; borrowers below $25k default at about 2.46× the portfolio rate. **So what**: Review affordability and existing obligations, and monitor this segment more closely.
3. **Employment:** Unemployed borrowers have a 34.96% default rate across 718 accounts, versus 14.97% for salaried borrowers. **So what:** Assess income stability and repayment capacity alongside other risk indicators—not as a standalone rule.

Assumptions, limitations and monitoring

The dataset and labels are synthetic, so results may not generalize to real lending; findings are associative, not causal. Assume labels and features are accurate and consistently defined. Confirm all inputs exist at scoring time—post-origination repayment or collection data must not enter an origination model. The decision threshold should be validated against business costs and review capacity.

Before production, validate bank data definitions and point-in-time availability; implement secure deployment/API integration, governed access, role-based controls and audit logs. Monitor data quality, missing values, feature and score drift, and—once outcomes mature—ROC-AUC, PR-AUC, KS, Gini, precision, recall, calibration and fairness across relevant groups. Define alert thresholds, review owners, and procedures for investigation, rollback, retraining and revalidation.